

STATE OF COLORADO

DIVISION OF MINERALS AND GEOLOGY
Department of Natural Resources

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March 6, 2003

Mr. Steve Mossberg
CEMEX
P.O. Box 529
Lyons, CO 80540

RE: CEMEX, Inc., Lyons Mine, Permit No. M-1977-208
Reclamation Cost for Demolition of the Cement Plant

Dear Mr. Mossberg,

The Division received the initial bond calculation for the Lyons Mine, Permit No. M-1977-208 conducted by CEMEX's consultant Banks and Gesso. The Division has the following comments regarding the estimate that will need to be addressed by the operator and incorporated into the reclamation cost estimate;

1. Banks and Gesso states that equipment, steel structures, conveyors and offices on the site have salvage values greater than the cost of removal. Therefore, the cost of equipment removal or demolition was not included in the bond calculation. The Division concurs that mining equipment and machinery stored on site does not need to be included in the reclamation cost estimate (and any contents within the buildings). However, structures such as offices, steel structures (gas tanks, utility lines, etc..) and conveyors need to be included in the reclamation cost estimate. The Division does not accept the face value of salvage credit for structures unless it meets the standards defined by Rule 4.12 of the Construction Materials Rules and Regulations.

Rule 4.12 specifically outlines the requirements for salvage credit. In order to determine the appropriate salvage credit for each structure, CEMEX will be required to provide an itemized listing of the appraised salvage value of each structure, not including the salvage value of fixtures contained therein (4.12.2(1)). The appraisal should be by a qualified independent appraiser (4.9.1(2)(b)(i) and 4.12.2(3)). CEMEX will also need to annually verify the value of the structure (4.12.2(1)) and to provide an annual fee to fund the Division's cost to establish and review the salvage value component of the bond (4.1.2(9)). In addition CEMEX would have to report on encumbrances (4.12.2(2)) and would not be allowed to remove the structure from the permit area without prior consent of the Board (4.12.1(c) and 4.9.3(c)) and would be required to insure the structures (4.12.1(c) and 4.9.3(c)). A copy of the regulations regarding salvage credit is included for reference.

In addition, 100 percent of the salvage value could not be deducted from the amount of the required bond. Rather, some percentage of the total salvage credit could be deducted to allow for contractor profit and for administrative costs. Rule 4.9.1(1) states that the Board or Office may accept interests in real or personal property as Financial Warranties of not

more than 85 percent of the estimated value of such property. Hence, the estimated salvage value minus the costs for contractor profit and administrative costs, would be further reduced by a factor of .85 before being applied to the amount of required bond.

CEMEX will need to comply with the requirements of Rule 4.12 if salvage credit is used to off set costs within the reclamation cost estimate. Without full compliance, the cost to remove these structures will need to be included in the estimate.

2. Banks and Gesso presented two estimates for demolishing the plant area. Cost Estimate A assumes that all concrete is crushed to various sizes, and that the material is sold as aggregate for a total cost of \$3.9 million. Cost Estimate B includes a total cost of the demolition and reclamation assuming that all material is demolished, stockpiled and covered and the area is revegetated. Cost Estimate B totals \$4.6 million. CEMEX has agreed to submit \$3.9 million, which reflects the cost associated with crushing and selling the material as an aggregate product. Please be aware that the intent of the financial warranty is to insure that the State of Colorado has the necessary funds to complete reclamation in accordance with the approved mining and reclamation plan in the event the bond is forfeited. The Division is unable to accept an estimate that requires the State of Colorado to process material to be sold as aggregate, thus in turn acting as a mine operator. The Division's jurisdiction is to reclaim the site to the standards set forth in the permit, and ensure that the financial warranty can achieve this goal.

Cost Estimate B includes a cost for demolition and disposal of the material on site in the amount of \$4.6 million. The Division believes this scenario should be applied to reflect the required financial warranty for the Lyons site. However, in order to approve this plan, some modification to the mining and reclamation plan may be required. This is discussed in more detail below.

3. *In order for mining structures and debris to be disposed of within the permit area, the operator will need to revise the mining and reclamation plan. Specifically, the operator will need to revise the permit to designate a disposal location for the structures and concrete to be buried on site. In addition, the operator will need to commit to a cover depth, and the source for the cover material (including a cover depth for all foundations to remain in place). Without such designation, the Division will need to assume that all material is hauled and disposed of in an approved landfill off site. The cost for such disposal in an off site landfill is much higher than the cost proposed by Banks and Gesso.*

The Division recently approved such a technical revision for the Henderson Mine and Mill (Permit No. M-1977-342) in order to reduce the structural demolition cost. Within the submittal, the operator designated a location for the debris, committed to 3' of overburden material over the disposal site and included information to protect the surface and groundwater by lining the disposal site with an impermeable material. The technical revision filed by Henderson substantially reduced the cost for demolishing structures at the mine.

If CEMEX proposes to dispose of the material within the affected area of the permit, then a technical revision to the permit would be required. In addition, the components of the reclamation cost estimate prepared by Banks and Gesso should also be included in the technical revision.

4. Banks and Gesso applied a unit cost of \$0.25/cf to demolish all buildings and structures on site. The Division has a more detailed unit cost for structural demolition depending on the composition of the structure, height and disposal location (within 50 feet or hauled to a landfill). If CEMEX proposes to demolish and bury structures on site, the cost could range from \$0.115 to \$0.150, less than the \$0.25 proposed by Banks and Gesso. However, if the structures need to be hauled to an approved landfill off site, the cost may range between \$0.694 and \$0.837, considerably higher than the Banks and Gesso estimate. Attached is a table of demolition costs for your reference.
5. CEMEX provided a bond calculation map that is very helpful in determining the nature and location of structures at the cement plant. However, the map did not contain a scale necessary to determine the size of the structures. Please notify the Division as to the scale of the map, or provide a new map depicting a scale.
6. On Table 2 of Estimate B, Banks and Gesso itemized the quantity and unit cost of concrete demolition in the amount of 2,698 cy for the grinding process area and 3,768 cy for the burning process area. However, Table 4 and Table 5 outlines the "concrete in place" for each structure (16,828 cy in the grinding area and 10,068 cy for the burning area), which is significantly higher than the amounts listed in Table 2. Please explain why the "concrete in place" amount was not used as a factor in the reclamation cost estimate.
7. Please itemize all hazardous waste and chemicals stored on site that will need to be removed upon reclamation.
8. The reclamation cost estimate needs to include additional factors that would be incurred by the Division in the event of bond forfeiture. A cost for mobilization/demobilization to and from the job site should be included in the estimate. In addition, a number of factors make up the indirect costs and include liability insurance (1.55% of direct), performance bond (1.05% of direct), profit (10% of direct), engineering work and/or contract/bid preparation work, and reclamation management and/or administration (5% of direct and mandated by Rule 6.3.4(2)(c)). Please incorporate these costs into the reclamation cost estimate.

Please address the above noted questions/concerns by April 11, 2003. If you have any questions, please do not hesitate to contact me at 303-866-4943.

Sincerely,



Erica S. Crosby
Environmental Protection Specialist

Enclosure(s)

cc: Carl Mount; DMG